

main.c



Run

Output

```
1 #include <stdio.h>
2 #include <pthread.h>
3 #include <semaphore.h>
4 sem_t forks[5];
5 void *eat(void *arg)
6 {
7     int i = *(int *)arg;
8     printf("Philosopher %d is thinking\n", i + 1);
9     sem_wait(&forks[i]);
10    sem_wait(&forks[(i + 1) % 5]);
11    printf("Philosopher %d is eating\n", i + 1);
12    sem_post(&forks[i]);
13    sem_post(&forks[(i + 1) % 5]);
14    return NULL;
15 }
16 int main()
17 {
18     pthread_t p[5];
19     int i, id[5];
20     for(i = 0; i < 5; i++)
21         sem_init(&forks[i], 0, 1);
22     for(i = 0; i < 5; i++)
```

Philosopher 1 is thinking
Philosopher 1 is eating
Philosopher 3 is thinking
Philosopher 2 is thinking
Philosopher 3 is eating
Philosopher 5 is thinking
Philosopher 5 is eating
Philosopher 2 is eating
Philosopher 4 is thinking
Philosopher 4 is eating

=== Code Execution Successful ===